

Data Visualization and Exploration

COMS7056A/COMS4060A

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Lecture 1



Lesson Plan

- Introduction
- General Information
- Course Outline
- Desired outcomes
- Assessment
- Motivation

Introduction

- PhD Candidate at Wits and NYU.
- Research interest mainly in Evolutionary Algorithms, Reinforcement Learning and AI in games.
- Worked in startups and corporate industry.



General Information

- You can contact me at: muhammad.nasir@wits.ac.za
- Tutors: TBA
- Moodle will be used for all content.
- Lectures on Fridays from 15h00 to 17h00.
- Lectures will usually be 1 hour 30 minutes long.
- Office hours on Wednesdays from 09h00 to 12h00
- 2 assignments + 2 quizzes for 60%.
- Final exam for 40%.
- This course will be far less mathematical than other courses.

Course Outline (Subject To Change)

- Course Introduction
- Describing data with statistics, plots and other visualisations
- Feature engineering and data augmentation
- Data cleaning and outlier detection
- Feature selection
- Linear dimensionality reduction

Course Outline (Subject To Change)

- Non-linear dimensionality reduction
- Clustering techniques
- Metric learning and similarity
- Geospatial data
- Time series data (stationarity, autocorrelation, seasonality, FT, spectrograms)
- Communication with visualisation, dashboarding and notebooks

Desired Outcomes

- Introduction to common frameworks.
- Get an intuition of how you should visualise the data.
- Get a gist of how industry explores data.
- Use these techniques in your research papers.

Suggested Textbooks

- Textbooks: There isn't a specific textbook, but some good references include:
- Elements of Statistical Learning (T. Hastie, R. Tibshirani, J. Friedman)
- Data Mining: Practical Machine Learning Tools and Techniques (I. Witten, E. Frank, M. Hall)
- Python for Data Analysis (W. McKinney)
- Python Geospatial Analysis Cookbook (M. Diener)
- Practical Statistics for Data Scientists (P. Bruce, A. Bruce)
- Data Mining and Statistics for Decision Making (S. Tuffery)



Other Resources

- towardsdatascience.com
- seaborn.pydata.com
- datacamp.com
- scikit-learn.org
- analyticsvidya.com
- Python Data Science Handbook
(<https://jakevdp.github.io/PythonDataScienceHandbook/>)
- Additional source materials and references will be provided throughout



Motivation

- Increasingly living in a data-driven world.
- Data is collected in bulk.
- Need to make sense out of all of this data.
- Machine learning, AI, LLMs are the big craze right now.
- Know it before you “machine learn” it.
- 90% of your job will be cleaning data and exploring it.

Why We Do Data Exploration?

- What is Machine Learning?
 - A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .
 - Machine learning is the study of computer algorithms that improve automatically through experience.
 - *Tom Mitchell*
- Can you throw any data at your favorite LLM?
 - No.
- Maybe I want to cluster my data but the data has no natural clusters.
- Maybe one dimension is spread out and would end up dominating my clustering algorithm.

Why We Do Data Exploration?

- Are there any outliers?
- Missing values?
- How will these affect our algorithm?
- Any statistical significance?

Why We Do Data Visualisation?

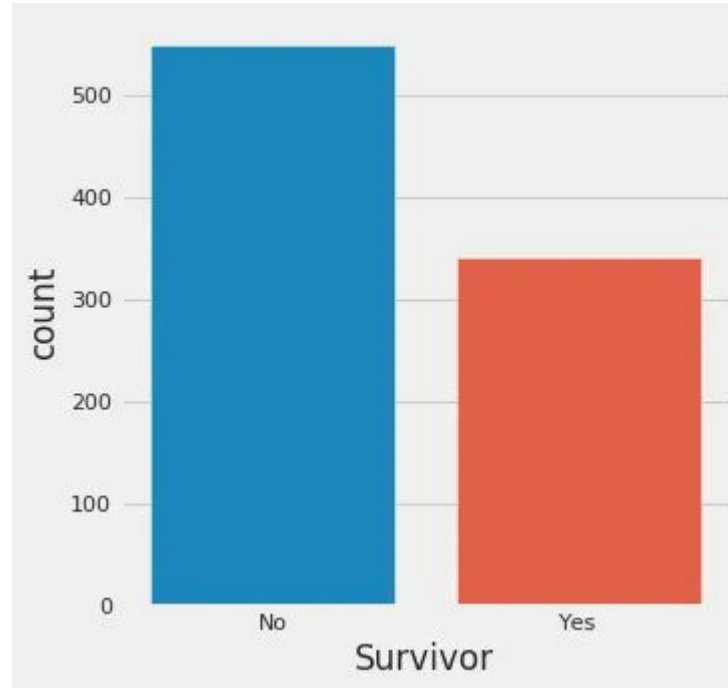
- As data grows in size it can be difficult to comprehend.
- We want to see visual patterns.
 - To understand different variables
 - To know what is actually predictable.
- Higher dimensional data into lower dimensional plots.
- Goal is to make sense out of it, to motivate, and capture ideas

Example

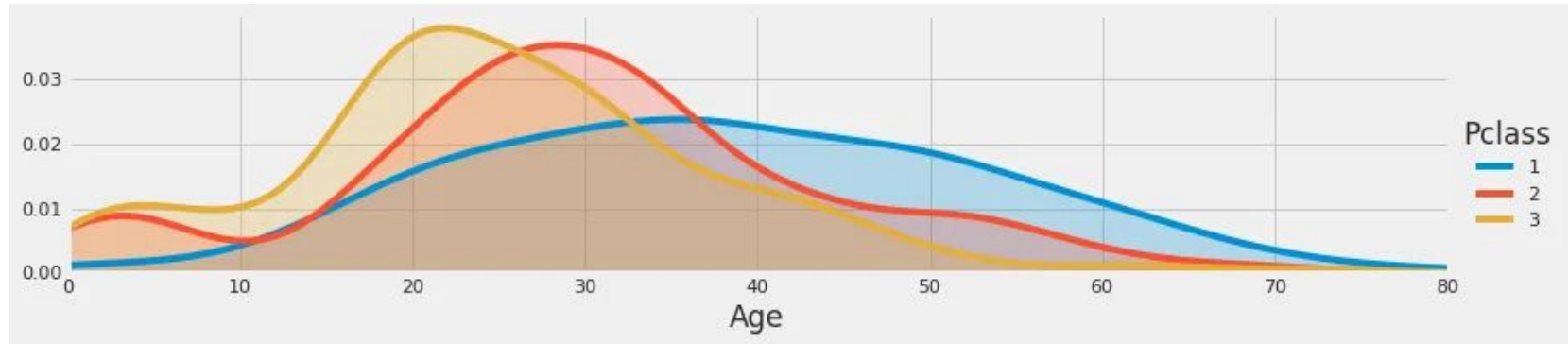
- Kaggle datasets are not real life datasets!
(<https://www.kaggle.com/competitions/titanic>)

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

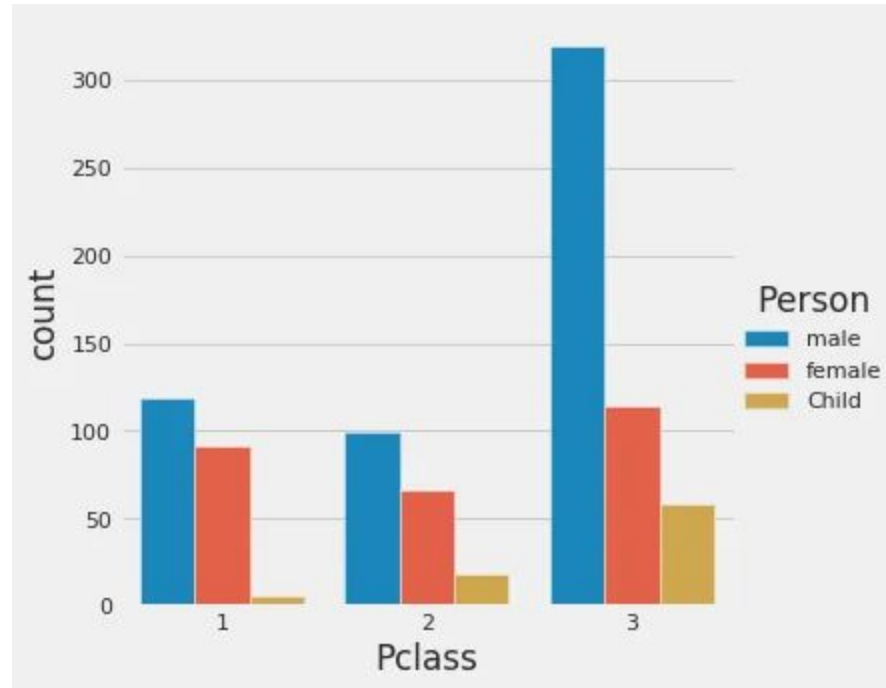
Example



Example



Example



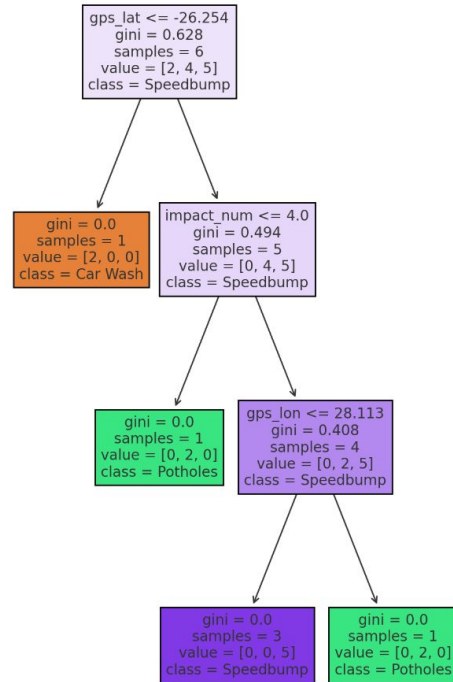
Example

- Real life datasets are messy!

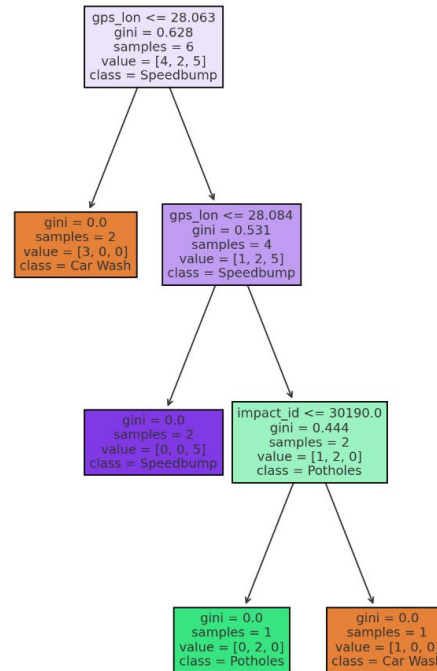
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1	8035	Potholes	-26.21976800601239	28.007901822081767	2018-02-14 11:27:10	0
2	42523	Potholes	-26.117395442219184	28.14589672507164	2018-07-23 16:02:46	1
3	43327	Speedbump	-26.036635507831715	28.079295523091076	2018-07-26 08:30:56	2
4	25785	Potholes	-26.170751300045556	28.088063508263502	2018-05-16 15:56:20	3
5	3704	Car Wash	-26.006330986618945	28.115868123977545	2018-01-24 14:46:19	4
6	39222	Speedbump	-26.146725061747016	28.059371930737672	2018-07-12 07:18:36	5
7	287	Car Wash	-26.269109487649303	28.048230625927008	2018-01-03 11:10:29	6
8	18137	Speedbump	-26.10229909587107	28.073136987310782	2018-04-11 17:41:45	7
9	24599	Speedbump	-26.238900179871735	28.027222031008748	2018-05-11 07:52:20	8
10	36303	Potholes	-26.091542830074115	27.969801979473907	2018-07-01 08:14:43	9
11	19227	Potholes	-26.132307493017912	28.147100216452642	2018-04-16 07:29:11	10
12	47644	Car Wash	-26.175608741797262	27.980711927031308	2018-08-09 15:10:36	11
13	34595	Car Wash	-26.202244674853148	28.113806886900875	2018-06-24 14:19:52	12
14	33889	Car Wash	-26.16550961396287	28.05312330800413	2018-06-21 10:12:03	13

Example

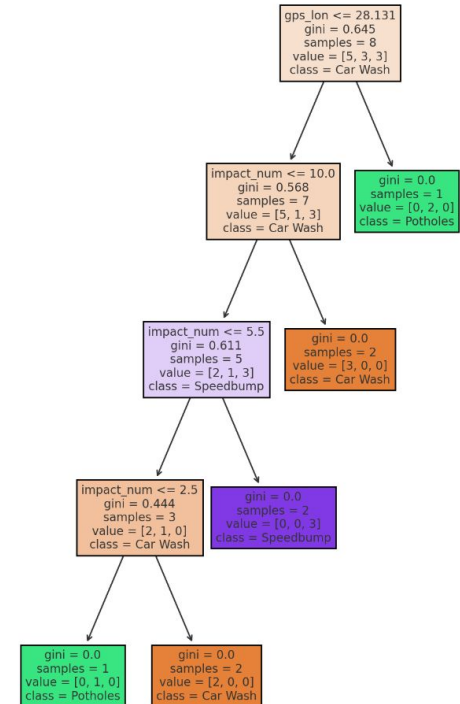
Tree 1



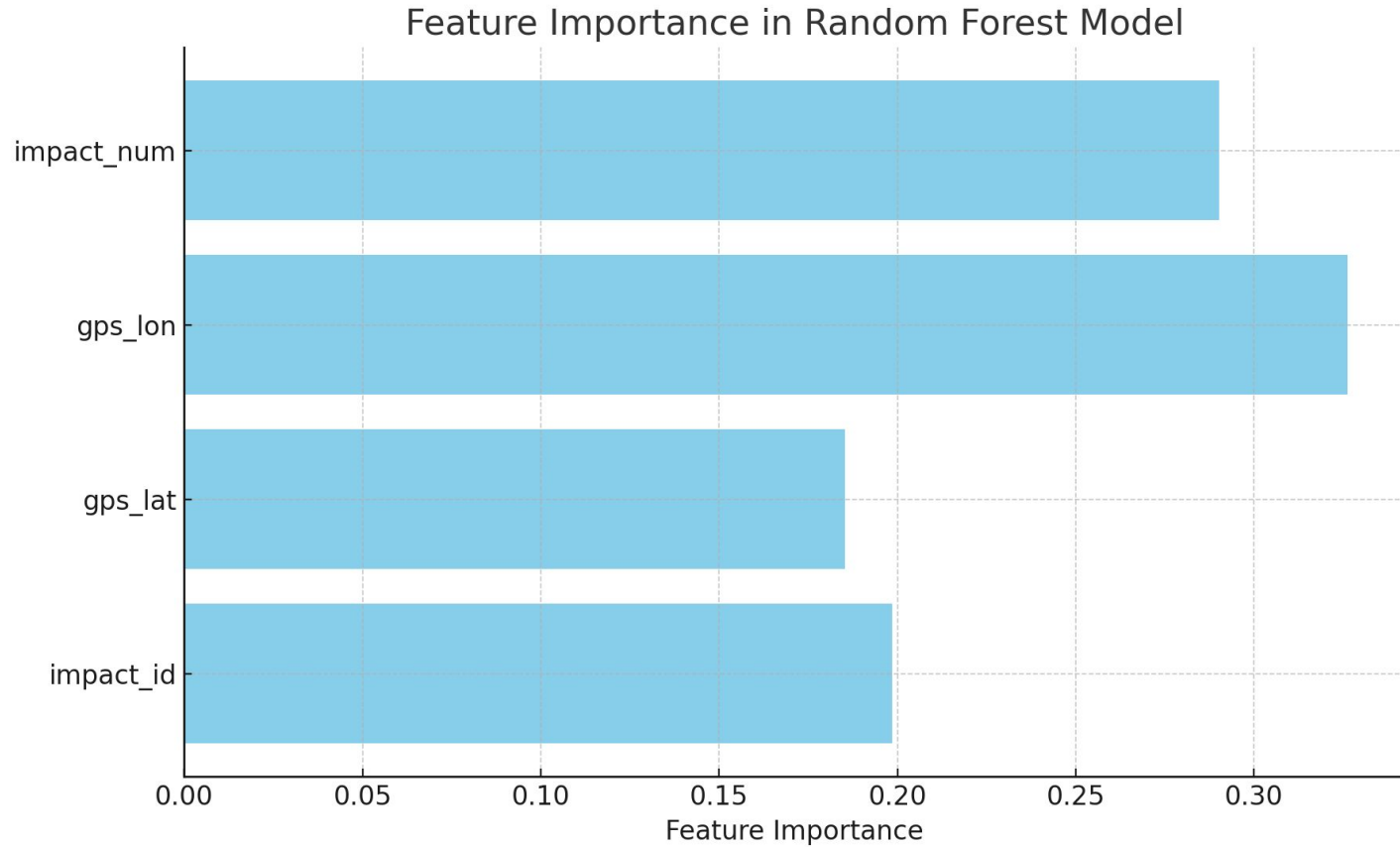
Tree 2



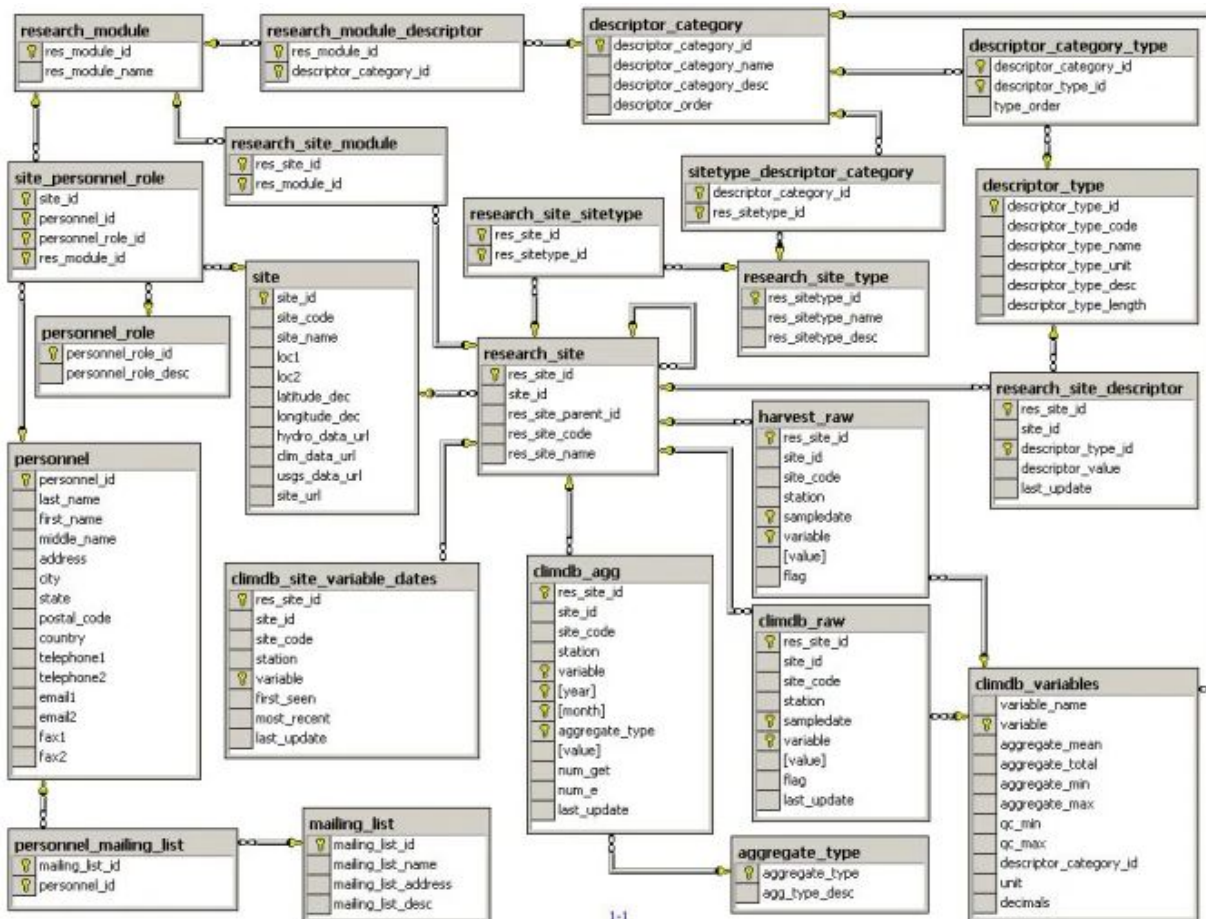
Tree 3



Example



Example



Curse Of Dimensionality

- Some data sets have thousands of dimensions.
- It become difficult to visualize.
- It becomes easy to overfit.
- We'll look at dimensionality reduction techniques to make it easier to visual data.



Conclusion

- Introduction
- Motivation
- Some examples.

